

Comprehensive Analysis of Field Oriented Control (FOC) for a 3-Phase Induction Motor

*Project Report Submitted in partial fulfilment of the requirements
for the degree of
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(formerly known as West Bengal University of Technology)*

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Certificate of Recommendation

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List of Principal Symbols and Acronyms

- **FOC:** Field Oriented Control
- **IFOC:** Indirect Field Oriented Control
- **SVPWM:** Space Vector Pulse Width Modulation
- **PI:** Proportional-Integral
- V_s : Stator Voltage
- i_{ds} : Direct-axis stator current (Flux-producing)
- i_{qs} : Quadrature-axis stator current (Torque-producing)
- ψ_r : Rotor flux linkage
- T_e : Electromagnetic Torque
- θ_e : Electrical rotor flux angle
- ω_m : Mechanical rotor speed
- ω_{sl} : Slip frequency

Abstract

This report provides a component-level technical analysis of the Simulink model VECTOR_CONTROL_FOC_OF_IM.slx, which implements an Indirect Field Oriented Control (IFOC) architecture for a 50 HP Squirrel-Cage Induction Motor.

Intended as an instructional document for electrical machines and drives coursework, this report deconstructs the theoretical and practical frameworks within the simulation. It establishes the plant's equivalent circuit parameters and dynamic modelling. The analysis explores the coordinate transformations (Forward and Reverse Clarke and Park) required to map stationary three-phase variables into a synchronously rotating $d - q$ reference frame.

By examining the decoupling equations, the report demonstrates how IFOC achieves independent regulation of rotor flux and electromagnetic torque, mirroring the dynamics of a separately excited DC machine. Furthermore, the document evaluates the discrete-time closed-loop control strategy, detailing the tuning parameters and anti-windup mechanisms of the cascaded Proportional-Integral (PI) controllers used in the outer speed and inner current loops. Finally, the report analyses the Space Vector Pulse Width Modulation (SVPWM) synthesis used to drive the three-phase inverter, noting its effect on DC bus utilisation and total harmonic distortion (THD).

This analysis connects electromechanical theory with digital control implementation.

Chapter 1

INTRODUCTION

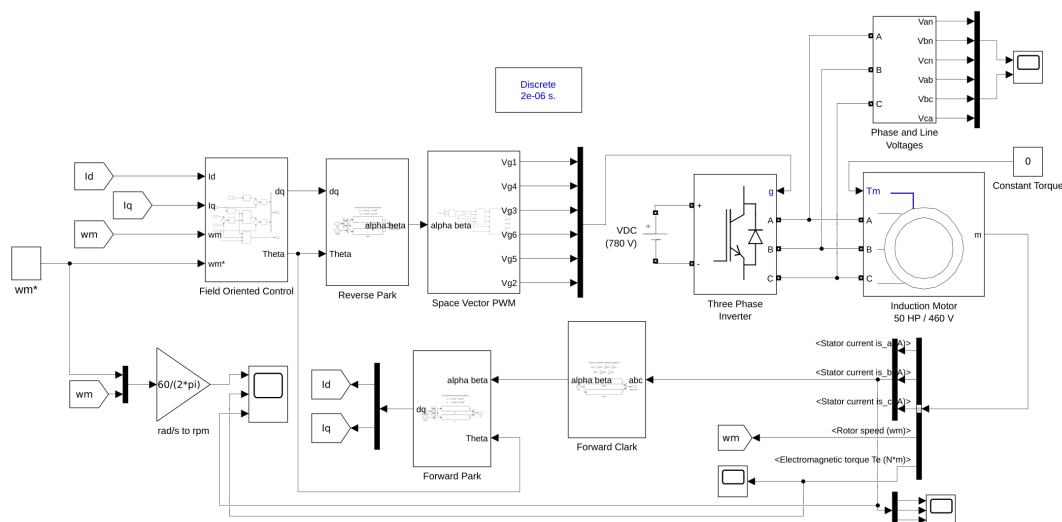


Figure 1.1: Overall Simulink Architecture of Field Oriented Control of IM

1.1 Introduction

Traditionally, the control of induction machines was dominated by scalar methods, such as constant Volts-per-Hertz (V/f) control. Scalar control governs the magnitude and frequency of the applied voltage and currents, independent of the spatial orientation of the electromagnetic fields. Consequently, scalar methods exhibit delayed transient responses, cross-coupling effects, and an inability to independently regulate torque and flux during dynamic state changes.

Field Oriented Control (FOC), also known as Vector Control, is an AC machine drive control method that addresses the limitations of scalar control. The objective of FOC is to emulate the decoupled control dynamics of a separately excited DC motor. In a traditional DC machine, the field flux and the armature flux are maintained in a state of spatial orthogonality (perpendicularity) by the mechanical commutator. This decoupling allows the field current and armature current to be manipulated independently to control magnetic flux and electromagnetic torque, respectively.

1.2 Review of Previous Works (Literature Review)

The evolution of AC motor control has progressed significantly. Early literature focused on addressing the coupled, non-linear multivariable nature of induction motors. Vector Control methodology introduced coordinate transformations (Clarke and Park) to project time-varying AC parameters into a synchronous, DC-like reference frame.

While Direct FOC techniques emerged first, relying on physical air-gap sensors for flux measurement, the industry moved toward Indirect Field Oriented Control (IFOC). IFOC mathematically estimates the rotor flux angle by compounding the measured rotor speed with a calculated slip frequency, providing a method used in industrial drives.

To replicate orthogonal decoupling in a three-phase AC induction machine, FOC treats the multi-phase stator currents as spatial vectors. Through mathematical coordinate transformations—namely the Clarke and Park transforms—the time-varying, three-phase AC variables (i_a, i_b, i_c) are projected onto a two-dimensional, synchronously rotating reference frame ($d - q$ axes). Because this reference frame rotates at the same synchronous speed as the machine's magnetic field, the sinusoidal AC quantities are transformed into DC steady-state values, allowing for the implementation of standard Proportional-Integral (PI) control loops.

By aligning the direct-axis (d -axis) of this rotating frame with the rotor flux linkage vector, the overall stator current is mathematically decomposed into two independent, orthogonal components:

Flux-Producing Component (i_{ds}): The direct-axis stator current is constrained by the control algorithm to align with the rotor flux linkage vector (ψ_r). In this oriented reference frame, i_{ds} acts analogously to the field excitation current (I_f) in a separately excited DC motor; it is responsible for establishing and maintaining the internal magnetisation of the induction machine.

Under steady-state conditions, the relationship is linear: the rotor flux magnitude is governed by $\psi_r = L_m i_{ds}$. In the constant-torque region of operation (below base speed), i_{ds} is maintained at a constant nominal value. Conversely, in the constant-power region (above base speed), i_{ds} is attenuated to facilitate “field weakening,” reducing the back-EMF to allow further speed increases within the inverter's voltage limit.

Torque-Producing Component (i_{qs}): The quadrature-axis stator current is maintained at a 90° electrical angle relative to the rotor flux axis. Because the machine's flux is isolated onto the d -axis ($\psi_{qr} = 0$), the cross-product interaction that generates mechanical force relies on this orthogonal q -axis current. It operates analogously to the armature current (I_a) in a DC motor.

The developed electromagnetic torque can be expressed via the decoupled equation:

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs} \quad (1.1)$$

[Calculation Execution] Substituting the motor parameters: $P = 4$, $L_m = 34.7 \text{ mH} = 0.0347 \text{ H}$, $L_r = L_{lr} + L_m = 0.8 \text{ mH} + 34.7 \text{ mH} = 35.5 \text{ mH} = 0.0355 \text{ H}$, and assuming nominal flux $\psi_r = 0.96 \text{ Wb}$:

$$T_e = \frac{3}{2} \left(\frac{4}{2} \right) \left(\frac{0.0347}{0.0355} \right) (0.96) \cdot i_{qs} \quad (1.2)$$

$$T_e = 3 \times (0.9775) \times (0.96) \cdot i_{qs} \quad (1.3)$$

$$T_e \approx 2.8152 \cdot i_{qs} \text{ N} \cdot \text{m} \quad (1.4)$$

This establishes a linear torque constant of $2.8152 \text{ N} \cdot \text{m/A}$ for the q -axis current. Because the rotor flux ψ_r is held constant by the independent d -axis control loop, the electromagnetic torque T_e is directly and linearly proportional to i_{qs} .

This spatial decoupling allows the induction motor to respond to step changes in torque demand, reducing the transient delay and cross-coupling dynamics of scalar (V/f) control methods.

A requirement for FOC is the real-time determination of the rotor flux angle (θ_e) to execute the Park and Inverse-Park transformations. The Simulink architecture analysed in this report utilises **Indirect Field Oriented Control (IFOC)**. Rather than relying on physical Hall-effect

sensors embedded in the air gap or state observers to directly measure the flux (Direct FOC), the IFOC scheme calculates the required flux angle. It achieves this by taking the measured mechanical rotor speed and adding it to a calculated, feedforward slip frequency.

1.3 Objective and Motivation

The primary objective of this project is to analyse and document a functional Simulink model of an IFOC system for a 3-phase induction motor. The motivation is to bridge theoretical electromechanical equations with discrete-time digital control logic. By deconstructing the model, the project demonstrates how continuous-time theory is translated into a simulated DSP environment for the regulation of torque and flux.

1.4 Scope of Present Work

The scope of this report covers the mathematical and functional breakdown of the VECTOR_CONTROL_FOC_OF_IM.slx Simulink model. This includes the identification of the 50 HP Squirrel-Cage Induction Motor plant parameters, the derivation of the $d - q$ decoupling equations, the tuning of the discrete PI control loops, and the execution logic of the Space Vector Pulse Width Modulation (SVPWM) algorithm. Hardware implementation is considered future scope and is beyond the bounds of this simulation-based analysis.

Chapter 2

INDUCTION MOTOR PLANT PARAMETERS AND DYNAMIC MODELLING

The electromechanical plant governed by this control architecture is a three-phase Squirrel-Cage Induction Motor (SCIM).

From a control perspective, the squirrel-cage design means the rotor currents and rotor flux linkages are physically inaccessible and cannot be directly measured. Consequently, the control strategy relies on a mathematical model of the machine, dictated by its equivalent circuit parameters.

Based on the model's initialisation routines and the machine block mask configurations, the motor is defined by the ratings and parameters summarised in Table 2.1.

Main	Callbacks	History	Description	Data
Model callbacks		Model initialization function:		
	PreLoadFcn			
	PostLoadFcn			
	InitFcn*			
	StartFcn			
	PauseFcn			
	ContinueFcn			
	StopFcn			
	PreSaveFcn			
	PostSaveFcn			
	CloseFcn			

```
clc;
clear all;
% Parameters of a induction machine.
Rs=0.12; %Stator resistance
Rr=0.02; %Rotor resistance
Lls=0.0001; %Stator inductance
Llr=0.0001; %Rotor inductance
Lm=0.01;
P=2; %Number of Poles
Wg=2*pi*50;
s=0.02; % slip
Wr=(1-s)*Wg;
```

Figure 2.1: Model Initialization Function (InitFcn) showing Plant Parameters

Table 2.1: Squirrel-Cage Induction Motor Nominal Ratings and Equivalent Circuit Parameters

Parameter Description	Symbol	Value
Nominal Ratings		
Nominal Power	S	50 HP (37.3 kW)
Stator Line-to-Line Voltage	V_s	460 V
Nominal Frequency	f	50 Hz
Number of Poles	P	4 (2 Pole Pairs)
Synchronous Speed	N_s	1500 RPM
Electrical Parameters (Per Phase)		
Stator Resistance	R_s	0.087 Ω
Rotor Resistance (Referred to Stator)	R_r	0.228 Ω
Stator Leakage Inductance	L_{ls}	0.1 mH
Rotor Leakage Inductance (Referred)	L_{lr}	0.8 mH
Magnetizing Inductance	L_m	34.7 mH
Mechanical Parameters		
Moment of Inertia	J	0.662 kg·m ²
Viscous Friction Coefficient	F	0.1 N·m·s/rad

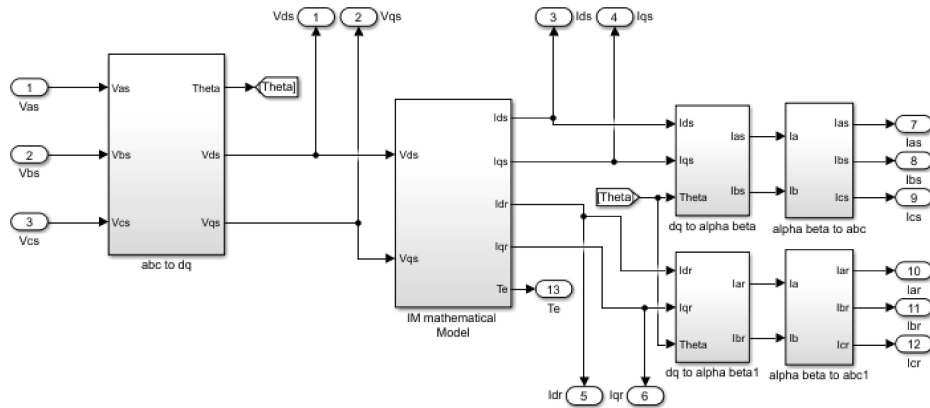


Figure 2.2: ABC to DQ Transformation and IM Mathematical Model Subsystem

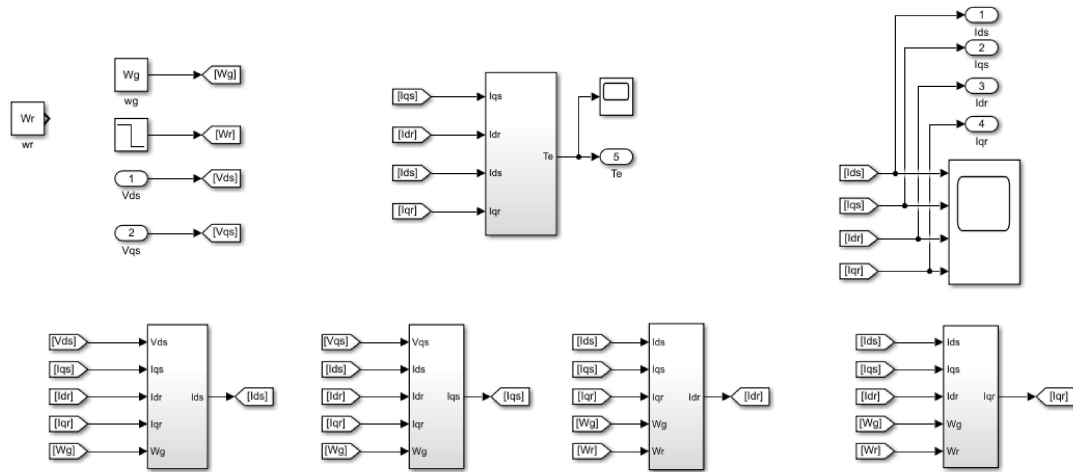


Figure 2.3: Internal Structure of the IM Mathematical Model

2.1 Significance of Electrical Parameters in FOC

In Vector Control, these parameters shape the dynamic transient response of the decoupling matrices. From the values provided, the lumped self-inductances of the stator (L_s) and rotor (L_r) are derived:

$$L_S = L_{lS} + L_m = 0.1 \text{ mH} + 34.7 \text{ mH} = 34.8 \text{ mH} \quad (2.1)$$

$$L_r = L_{lr} + L_m = 0.8 \text{ mH} + 34.7 \text{ mH} = 35.5 \text{ mH} \quad (2.2)$$

Converting the inductances to Henrys for base SI unit compatibility:

$$L_s = 34.8 \times 10^{-3} \text{ H} = 0.0348 \text{ H} \quad (2.3)$$

$$L_r = 35.5 \times 10^{-3} \text{ H} = 0.0355 \text{ H} \quad (2.4)$$

A required parameter in Indirect Field Oriented Control (IFOC) is the **rotor time constant** (τ_r), defined as:

$$\tau_r = \frac{L_r}{R_r} = \frac{35.5 \times 10^{-3}}{0.228} \approx 0.1557 \text{ seconds} \quad (2.5)$$

In the IFOC feedforward slip-calculation pathway, the slip frequency (ω_{sl}) is computed continuously as a function of the torque-producing current reference (i_{qs}^*) and this time constant. If the value of R_r deviates during operation (e.g., due to thermal heating of the rotor bars), τ_r will change, leading to **detuning**. Detuning causes spatial misalignment between the expected and actual $d - q$ axes, resulting in cross-coupling where torque commands perturb the machine flux.

The machine's **total leakage factor** (σ), which dictates the maximum rate of change of stator currents, is calculated as:

$$\sigma = 1 - \frac{L_m^2}{L_s L_r} \approx 1 - \frac{(34.7)^2}{(34.8)(35.5)} \approx 0.025 \quad (2.6)$$

This magnetic relationship ($\sigma \ll 1$) requires high-gain current loops to force the stator currents to track the FOC references.

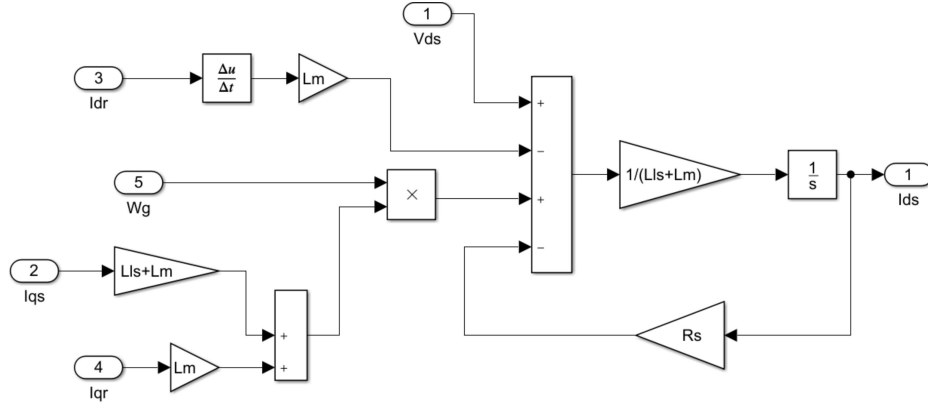


Figure 2.4: Dynamic State-Space Modeling of the d -axis Stator Current (I_{ds})

2.2 Mechanical Dynamics and Speed Loop Tuning

The mechanical parameters—the moment of inertia ($J = 0.662 \text{ kg} \cdot \text{m}^2$) and the viscous friction coefficient ($F = 0.1 \text{ N} \cdot \text{m} \cdot \text{s/rad}$)—govern the low-frequency dynamics of the plant. While the electrical dynamics operate on the order of milliseconds, the mechanical dynamics operate on the order of tenths of seconds or slower. This temporal separation is the justification for the cascaded control architecture in this model, where an outer speed loop provides the torque command (T_e^*) to the inner current loops.

The mechanical equation of motion that dictates the rotor's dynamic trajectory is governed by:

$$T_e - T_L = J \frac{d\omega_m}{dt} + F \omega_m \quad (2.7)$$

Substituting the mechanical parameters of the 50 HP motor:

$$T_e - T_L = 0.662 \frac{d\omega_m}{dt} + 0.1 \cdot \omega_m \quad (2.8)$$

where T_e is the developed electromagnetic torque, T_L is the applied load torque, and ω_m is the mechanical rotor speed in rad/s.

To tune the outer Proportional-Integral (PI) speed controller, the plant is analysed in the complex frequency domain. Assuming the load torque T_L acts as an independent disturbance ($T_L = 0$ for the primary plant transfer function), taking the Laplace transform of the mechanical equation yields:

$$\frac{\omega_m(s)}{T_e(s)} = \frac{1}{Js + F} = \frac{1}{0.662s + 0.1} \quad (2.9)$$

This first-order transfer function indicates the mechanical plant acts as a low-pass filter. The inertia of the 50 HP motor provides kinetic energy storage that attenuates high-frequency torque ripples generated by the discrete switching of the SVPWM inverter. However, this inertia imposes an upper limit on the achievable bandwidth of the speed control loop.

The outer speed PI controller must be tuned to address this pole:

- **Proportional Gain (K_p):** Provides the control effort to overcome the inertial mass (J) during transient speed changes.
- **Integral Gain (K_i):** Addresses steady-state tracking error and rejects steady-state load torque disturbances (T_L) applied to the shaft.

In practice, the speed loop bandwidth is constrained to be at least one decade slower than the inner q -axis current loop bandwidth. This frequency separation ensures that the inner loop appears as a gain of 1 from the perspective of the outer loop.

Chapter 3

CONTROL SYSTEM ARCHITECTURE AND DISCRETE-TIME EXECUTION

The Simulink model is architected around cascading subsystems that execute the Indirect Field Oriented Control (IFOC) algorithm. A parameter governing this digital control system is the fundamental sample time, set to $T_s = 2 \mu s$. This execution rate (equivalent to a 500 kHz update frequency) ensures that the discrete-time z-domain implementation introduces minimal phase delay. This temporal granularity is required for the synthesis of the Space Vector PWM signals and stability in the inner current regulation loops.

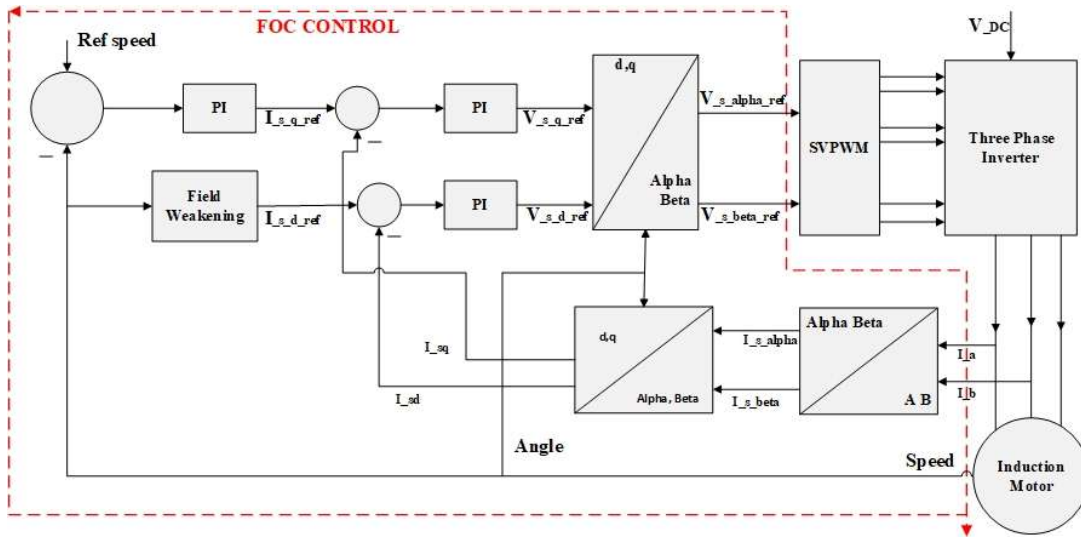


Figure 3.1: FOC Control Architecture Schematic

3.1 Coordinate Transformations: The Mathematical Foundation of FOC

FOC resolves the limitation of standard AC linear controllers: the inability of a standard Proportional-Integral (PI) controller to track a time-varying sinusoidal reference without a steady-state magnitude and phase error. By transforming the time-varying, three-phase AC stator variables into a synchronously rotating $d - q$ reference frame, FOC mathematically forces these alternating signals to appear as DC quantities. This allows PI controllers to achieve zero steady-state error.

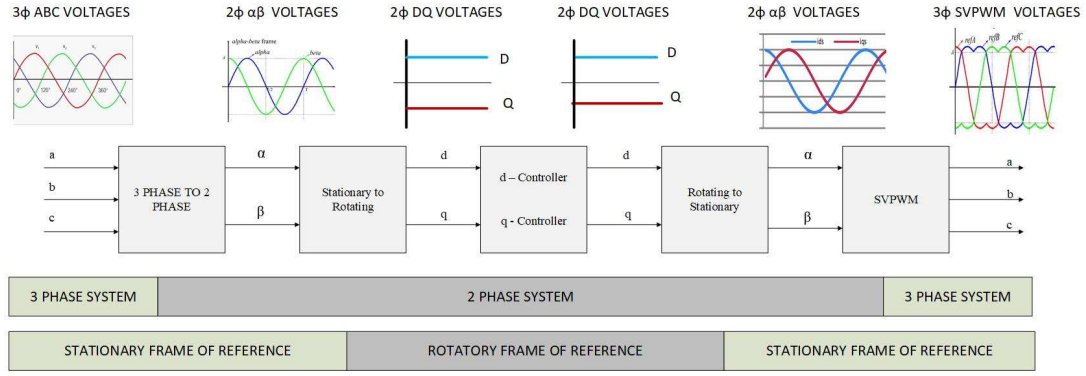


Figure 3.2: Overall Reference Frame Transformation Process

Forward Clarke Transform (Stationary 3 → 2 Projection): The initial stage of the vector control algorithm simplifies the three-phase AC system by projecting the measured stator phase currents (i_a, i_b, i_c) onto an equivalent two-dimensional orthogonal reference frame ($\alpha - \beta$ axes). In a star-connected induction motor without a neutral return path, Kirchhoff's current law dictates that the sum of the phase currents is zero ($i_a + i_b + i_c = 0$). Consequently, the zero-sequence homopolar component (i_0) is zero and discarded.

To ensure that the peak amplitude of the phase currents is preserved in the two-phase system, the model utilises the amplitude-invariant Clarke transformation matrix:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad (3.1)$$

Forward Clarke Transformation

$$\alpha(t) = \frac{2}{3}a(t) - \frac{1}{3}b(t) - \frac{1}{3}c(t)$$

$$\beta(t) = \frac{1}{\sqrt{3}}b(t) - \frac{1}{\sqrt{3}}c(t)$$

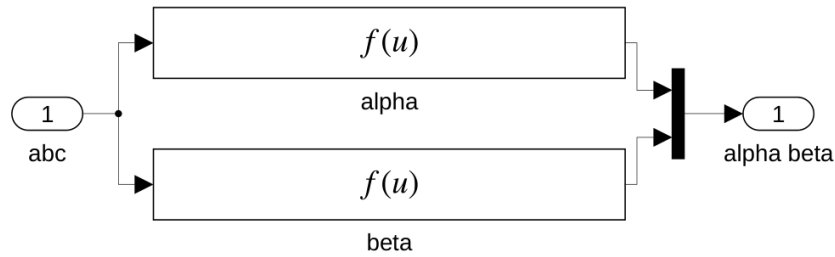


Figure 3.3: Forward Clarke Transformation Simulink Block

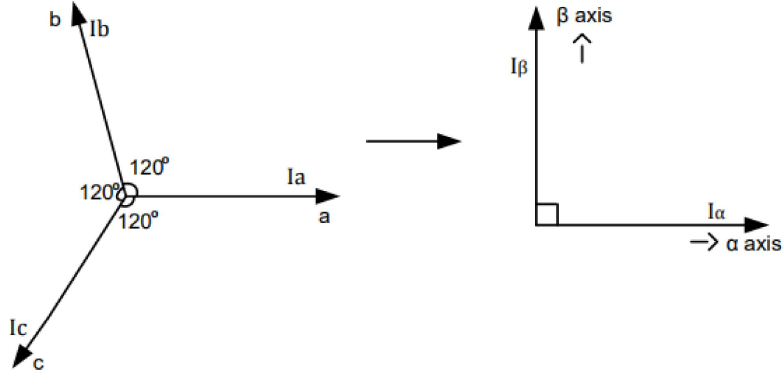


Figure 3.4: Vector Representation of the Clarke Transform

Assuming instantaneous balanced stator currents with a peak of 100 A, e.g., $i_a = 100\text{A}$, $i_b = -50\text{A}$, $i_c = -50\text{A}$ at $\omega t = 0$:

$$i_\alpha = \frac{2}{3} (1(100) - 0.5(-50) - 0.5(-50)) = 100 \text{ A} \quad (3.2)$$

$$i_\beta = \frac{2}{3} (0(100) + 0.866(-50) - 0.866(-50)) = 0 \text{ A} \quad (3.3)$$

The amplitude of 100 A is preserved in the $\alpha - \beta$ plane. While the resulting i_α and i_β components exist in an orthogonal complex plane, they remain time-varying sinusoidal waveforms oscillating at the fundamental stator frequency (ω_e).

Forward Park Transform (Stationary $\alpha - \beta$ to Rotating $d - q$ Projection): The Park transform resolves this control issue by projecting the stationary $\alpha - \beta$ frame onto a synchronously rotating reference frame ($d - q$ axes) that spins at the electrical angular velocity of the rotor magnetic flux. Utilising the instantaneous rotor flux angle (θ_e) synthesised by the IFOC slip estimator, the rotational mapping is defined as:

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \begin{bmatrix} \cos \theta_e & \sin \theta_e \\ -\sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} \quad (3.4)$$

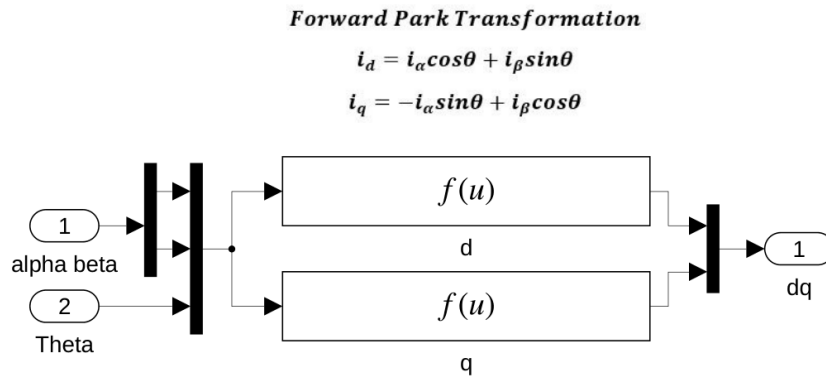


Figure 3.5: Forward Park Transformation Simulink Block

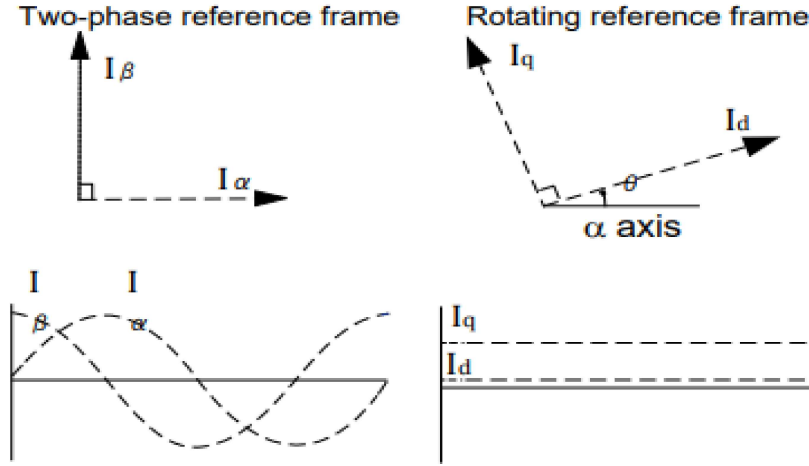


Figure 3.6: Vector Representation of the Park Transform (dq Rotating Frame)

From the perspective of this rotating frame, the fundamental AC currents are demodulated into DC values. By aligning the direct-axis (d) with the rotor flux linkage vector, i_d commands the magnetising flux, while the quadrature-axis current (i_q) independently commands the active electromagnetic torque.

Reverse Park Transform (Coordinate De-rotation for Actuation): Once the inner PI current regulators process the DC error signals in the $d - q$ domain, they generate the required decoupling voltage commands (v_d^* and v_q^*). Because the physical inverter and motor operate in the stationary reference frame, these rotating voltage vectors must be translated back. Since the Forward Park matrix is orthogonal, its inverse is its matrix transpose:

$$\begin{bmatrix} v_\alpha^* \\ v_\beta^* \end{bmatrix} = \begin{bmatrix} \cos \theta_e & -\sin \theta_e \\ \sin \theta_e & \cos \theta_e \end{bmatrix} \begin{bmatrix} v_d^* \\ v_q^* \end{bmatrix} \quad (3.5)$$

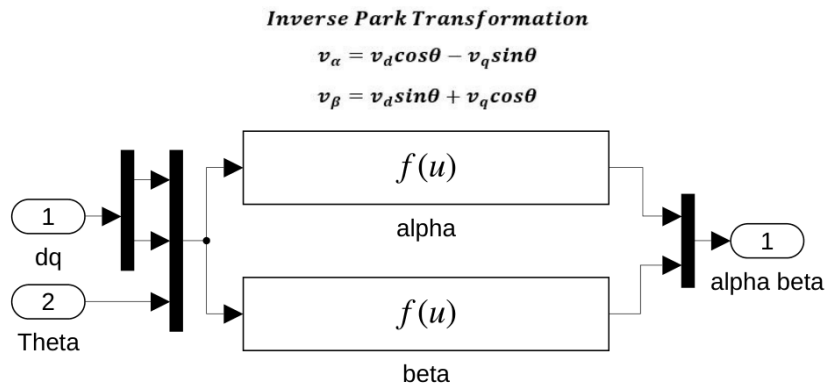


Figure 3.7: Inverse Park Transformation Simulink Block

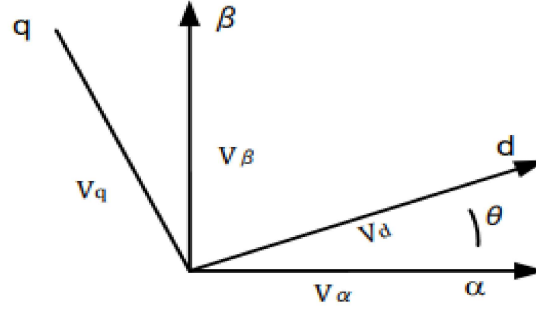


Figure 3.8: Vector Representation of the Inverse Park Transform

The resulting outputs, v_{α}^* and v_{β}^* , represent the commanded stator voltage space vector in the stationary plane. The control architecture terminates the reverse transformation sequence at this $\alpha - \beta$ stage because the downstream Space Vector PWM (SVPWM) subsystem is mathematically designed to accept the orthogonal v_{α}^* and v_{β}^* components.

3.2 Flux and Torque Decoupling Equations: The Core of IFOC

Within the Field Oriented Control subsystem, the decoupling equations translate the commanded electromagnetic torque (T_e^*) and the desired rotor flux (ψ_r^*) into their respective $d - q$ axis stator current references (i_{ds}^* and i_{qs}^*).

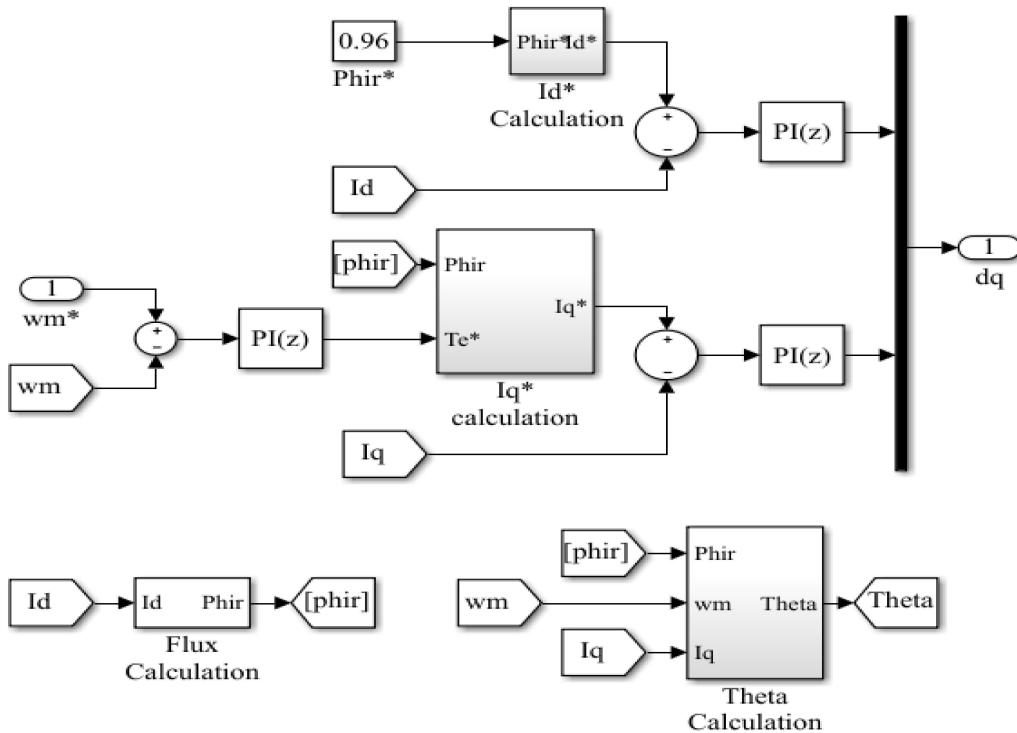


Figure 3.9: Detailed Internal Subsystems of the FOC Control Block

Direct-Axis Current Reference (i_{ds}^*): The direct-axis current is responsible for the magnetisation of the induction machine. The behaviour of the rotor flux linkage ψ_r is governed by the following first-order differential equation in the synchronous reference frame:

$$\tau_r \frac{d\psi_r}{dt} + \psi_r = L_m i_{ds} \quad (3.6)$$

Under a steady-state assumption ($\frac{d\psi_r}{dt} = 0$), the differential term is zero, and the reference direct-axis current calculation is:

$$i_{ds}^* = \frac{\psi_r^*}{L_m} \quad (3.7)$$

Reference I_d Calculation

$$I_d = \frac{\varphi_r}{L_m}$$

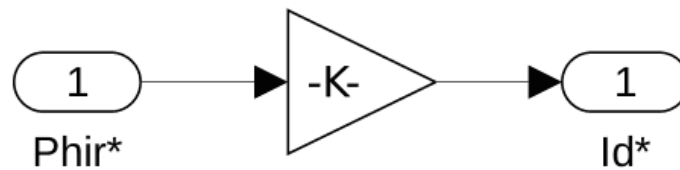


Figure 3.10: Reference I_d Calculation Block

Substituting $\psi_r^* = 0.96$ Wb and $L_m = 0.0347$ H:

$$i_{ds}^* = \frac{0.96}{0.0347} = 27.6657 \text{ A} \quad (3.8)$$

The d -axis current controller will regulate to 27.67 A to maintain nominal flux.

Quadrature-Axis Current Reference (i_{qs}^*): The quadrature-axis current controls the machine's developed torque. By aligning the d -axis perfectly with the rotor flux vector ($\psi_{dr} = \psi_r$, $\psi_{qr} = 0$), the torque equation simplifies to:

$$T_e = \frac{3}{2} \left(\frac{P}{2} \right) \frac{L_m}{L_r} \psi_r i_{qs} \quad (3.9)$$

To achieve a commanded torque T_e^* , we invert this relationship to solve for the necessary reference current i_{qs}^* :

$$i_{qs}^* = \frac{2}{3} \cdot \frac{2}{P} \cdot \frac{L_{lr} + L_m}{L_m} \cdot \frac{T_e^*}{\psi_r} \quad (3.10)$$

Reference I_q Calculation

$$T_e = \frac{3}{2} \frac{P}{2} \frac{L_m}{L_{lr} + L_m} I_q \varphi_r$$

$$I_q = \frac{2}{3} \frac{2}{P} \frac{L_{lr} + L_m}{L_m} \frac{T_e}{\varphi_r}$$

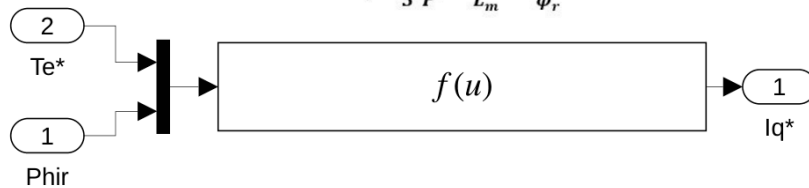


Figure 3.11: Reference I_q Calculation Block

Computational Safeguards in Digital Implementation: To maintain computational stability in the DSP/Simulink environment and avoid division-by-zero when flux $\psi_r \approx 0$ at startup, the calculation block introduces a numerical perturbation ($\varepsilon = 1 \times 10^{-3}$) to the denominator:

$$i_{qs}^* \propto \frac{T_e^*}{\psi_r + 10^{-3}} \quad (3.11)$$

This prevents infinite current commands during the transient window when the i_{ds} current is initially building the magnetic field.

3.3 Slip and Flux Angle Synthesis

The core of any IFOC scheme is synthesizing the flux angle θ_e mathematically in a feedforward manner. The necessary slip frequency (ω_{sl}) is computed as:

$$\omega_{sl} = \frac{L_m}{\tau_r} \cdot \frac{i_{qs}^*}{\psi_r^*} \quad (3.12)$$

To determine the synchronous speed of the reference frame (ω_e), this slip frequency is added to the electrical rotor speed ($\omega_r = \omega_m \frac{P}{2}$):

$$\omega_e = \omega_r + \omega_{sl} \quad (3.13)$$

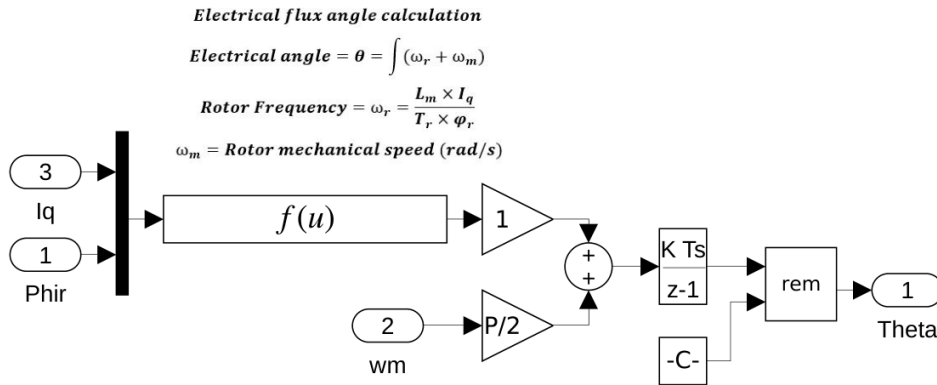


Figure 3.12: Electrical Flux Angle (θ) Calculation Block

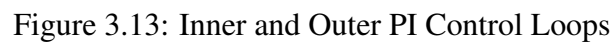
The instantaneous rotor flux angle θ_e is obtained by integrating the synchronous speed. In discrete-time:

$$\theta_e[k] = \left(\theta_e[k-1] + \omega_e[k] \cdot T_s \right) \mod 2\pi \quad (3.14)$$

The modulo operator wraps the angle within the $[0, 2\pi]$ interval to prevent numerical overflow in the DSP registers.

3.4 Discrete PI Controllers

The model utilises parallel-form discrete PI(D) controllers for closed-loop regulation.



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Chapter 4

SPACE VECTOR PULSE WIDTH MODULATION (SVPWM)

The control actuation in the Simulink model is executed by a Three-Phase Voltage Source Inverter (Universal Bridge), powered by a 780 V DC link. To translate the continuous domain v_{α}^* and v_{β}^* voltage commands into discrete transistor switching signals, the model employs a Space Vector PWM (SVPWM) subsystem.

SVPWM injects a zero-sequence (triplen harmonic) voltage into the phase commands, which prevents premature clipping and allows output to reach $\frac{V_{dc}}{\sqrt{3}}$. It schedules zero vectors to minimize ripple current, THD, and switching losses.

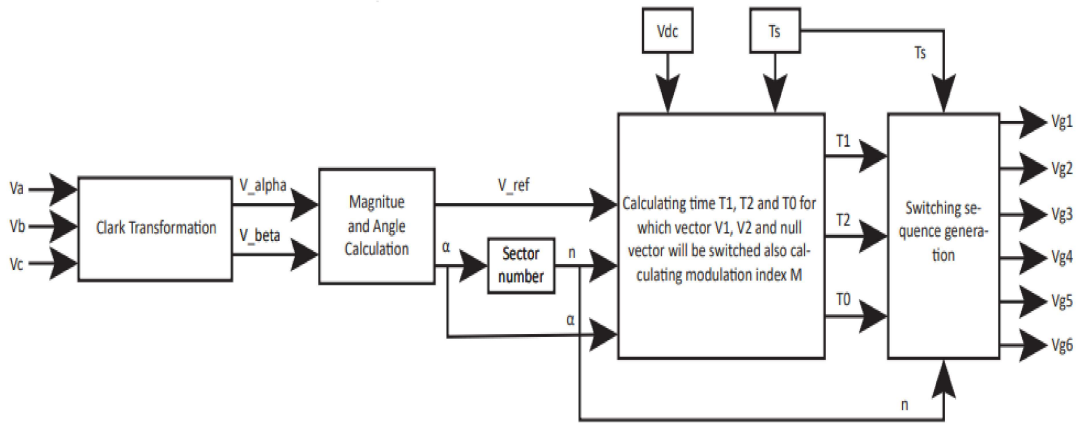


Figure 4.1: SVPWM Algorithmic Flowchart

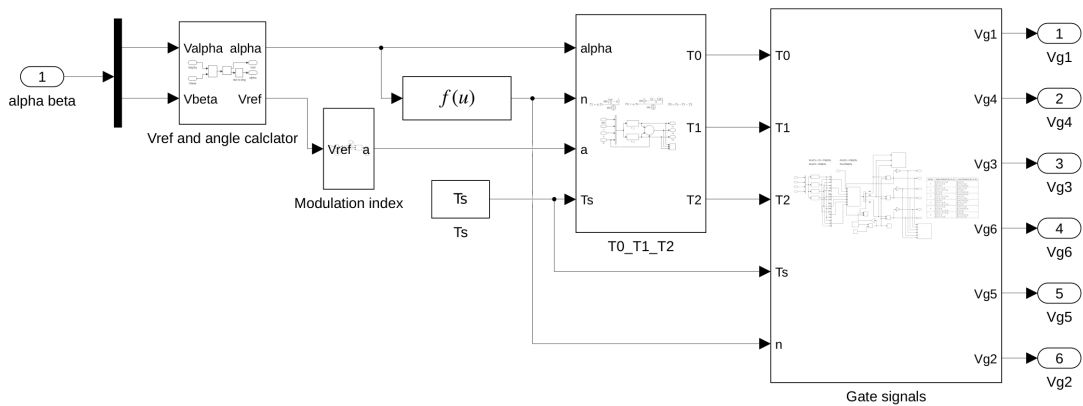


Figure 4.2: SVPWM Internal Subsystems in Simulink

4.1 SVPWM Algorithm Execution

Within the Space Vector PWM subsystem, the algorithm follows a sequence of mathematical operations during each $2\mu s$ sample period:

Reference Vector Synthesis: The incoming stationary frame voltage commands, v_α^* and v_β^* , define a desired reference voltage vector $\vec{V}_{ref}$. The model first computes the magnitude and angle of this vector:

$$|\vec{V}_{ref}| = \sqrt{(v_\alpha^*)^2 + (v_\beta^*)^2} \quad (4.1)$$

$$\theta_{ref} = \arctan\left(\frac{v_\beta^*}{v_\alpha^*}\right) \quad (4.2)$$

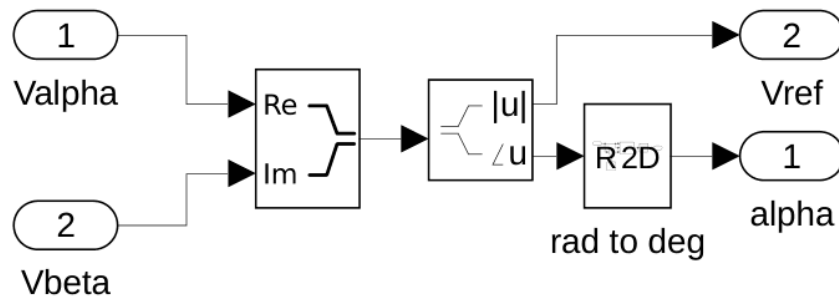


Figure 4.3: Vref Magnitude and Angle Calculation Block

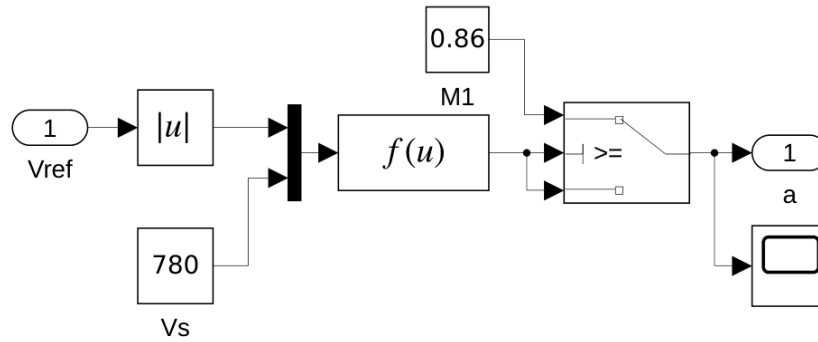


Figure 4.4: Modulation Index Safety Check Block

Dwell Time Calculation: Once the sector is identified, the SVPWM algorithm synthesises $\vec{V}_{ref}$ by time-averaging the adjacent active voltage vectors ($\vec{V}_1$ and $\vec{V}_2$) and the zero vectors over T_s . The dwell times are calculated as:

$$T_1 = \frac{\sqrt{3} \cdot T_s \cdot |\vec{V}_{ref}|}{V_{dc}} \sin\left(\frac{\pi}{3} - \theta'\right) \quad (4.3)$$

$$T_2 = \frac{\sqrt{3} \cdot T_s \cdot |\vec{V}_{ref}|}{V_{dc}} \sin(\theta') \quad (4.4)$$

$$T_0 = T_s - T_1 - T_2 \quad (4.5)$$

$$T1 = a.Ts \frac{\sin\left(\frac{n\pi}{3} - \alpha\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T2 = a.Ts \frac{\sin\left(\alpha - \frac{(n-1)\pi}{3}\right)}{\sin\left(\frac{\pi}{3}\right)} \quad T0 = Ts - T1 - T2$$

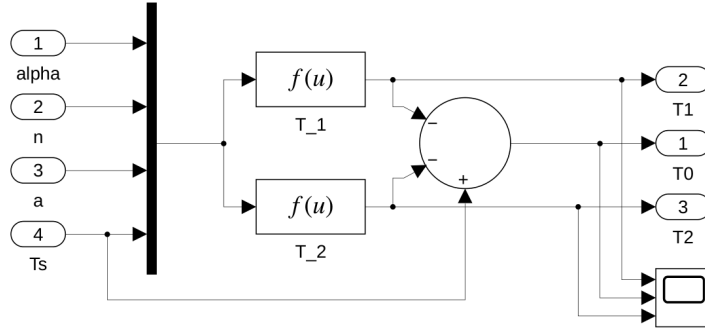


Figure 4.5: T1, T2, and T0 Calculation Subsystem

Switching Sequence Generation: Finally, the model translates these calculated dwell times into physical gate drive signals (V_{g1} through V_{g6}). A symmetrical switching pattern is employed (e.g., $\vec{V}_0 \rightarrow \vec{V}_1 \rightarrow \vec{V}_2 \rightarrow \vec{V}_7 \rightarrow \vec{V}_2 \rightarrow \vec{V}_1 \rightarrow \vec{V}_0$). The zero vector time T_0 is divided equally between the two null states.

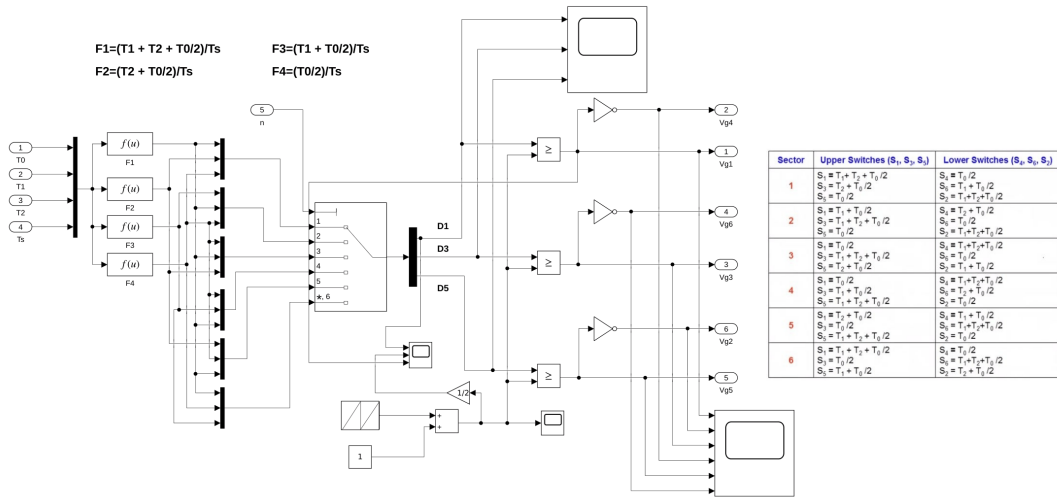


Figure 4.6: Sector Switching Sequence Generation Logic

Chapter 5

CONCLUSIONS AND FUTURE SCOPE

5.1 Conclusions

The provided Simulink model is a functional “digital twin” of an Indirect Field Oriented Control (IFOC) system for a Squirrel-Cage Induction Motor. By modelling the electromechanical plant, the coordinate transformations, and the discrete-time control logic, this architecture explains the non-linear dynamics inherent in three-phase AC machines.

By projecting the stator currents into a synchronously rotating $d - q$ reference frame, the architecture demonstrates that an induction motor can be controlled with the orthogonal precision of a separately excited DC motor. The simulation bridges continuous-time control theory and practical digital implementation operating at a $2\mu s$ sample time.

System Simulation Results

To validate the performance of the controller, several waveforms from the simulation are presented below:

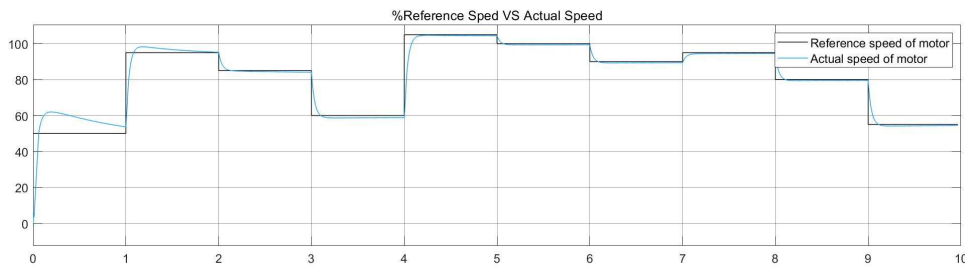


Figure 5.1: Reference vs Actual Speed Tracking Performance

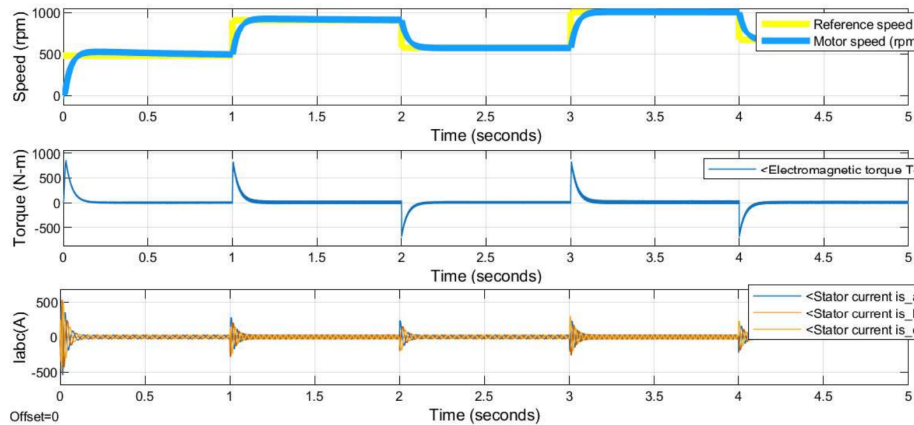


Figure 5.2: Dynamic Response of Motor Speed, Torque, and Phase Currents

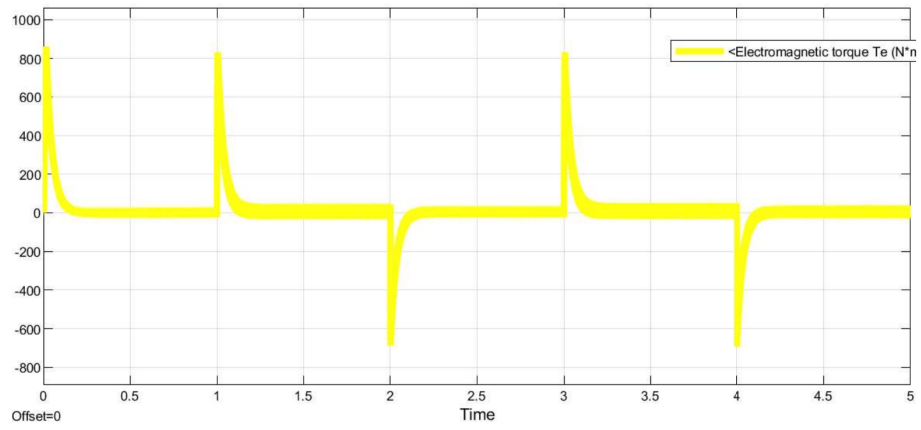


Figure 5.3: Electromagnetic Torque Transient Spikes during Acceleration/Deceleration

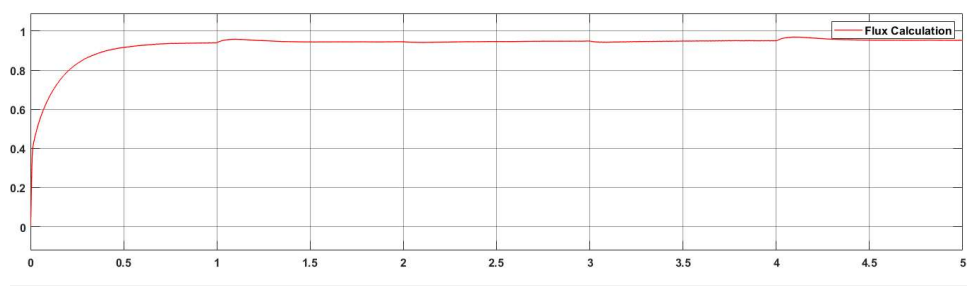


Figure 5.4: Rotor Flux Estimation Reaching Steady State

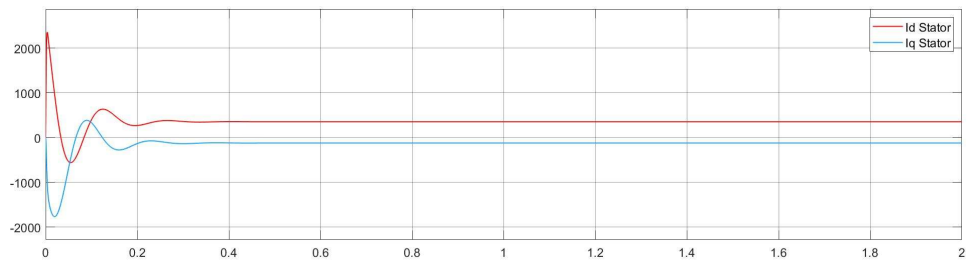


Figure 5.5: Stator d -axis and q -axis Currents (I_{ds} and I_{qs})

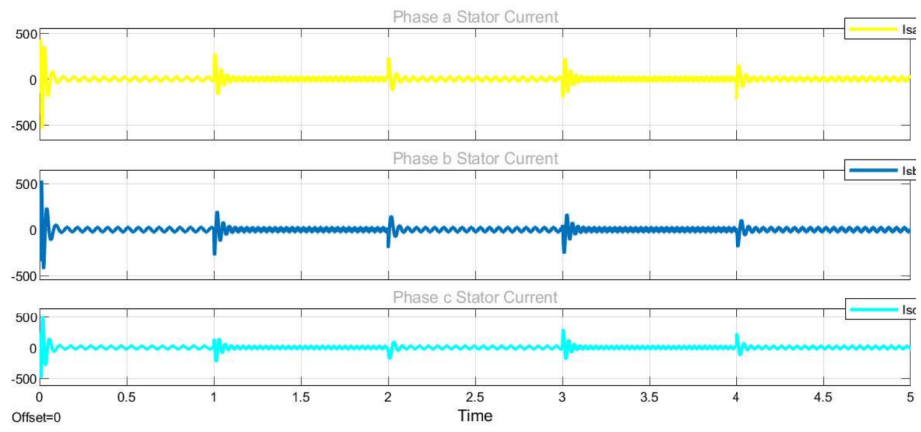


Figure 5.6: Three-Phase Stator Currents Under Dynamic Load Changes

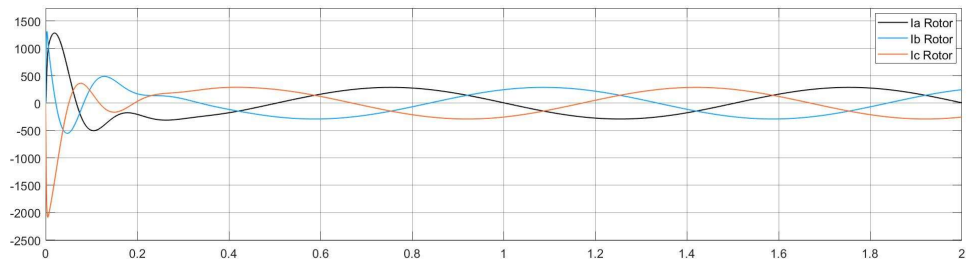


Figure 5.7: Induced Rotor Currents at Startup

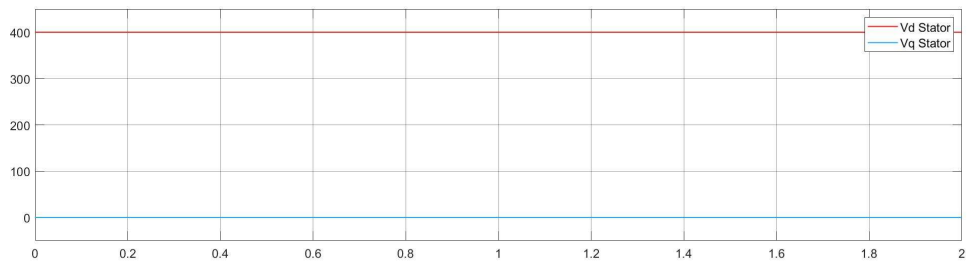


Figure 5.8: Steady-State Stator Voltages V_d and V_q

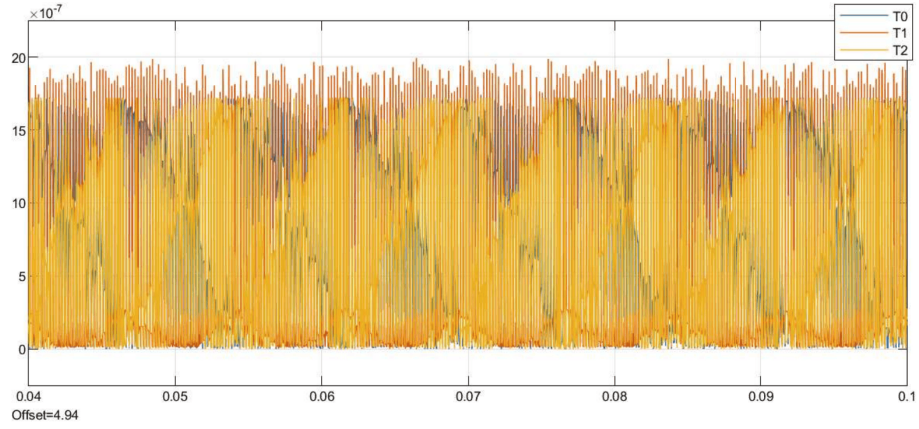


Figure 5.9: Calculated SVPWM Dwell Times (T_0, T_1, T_2)

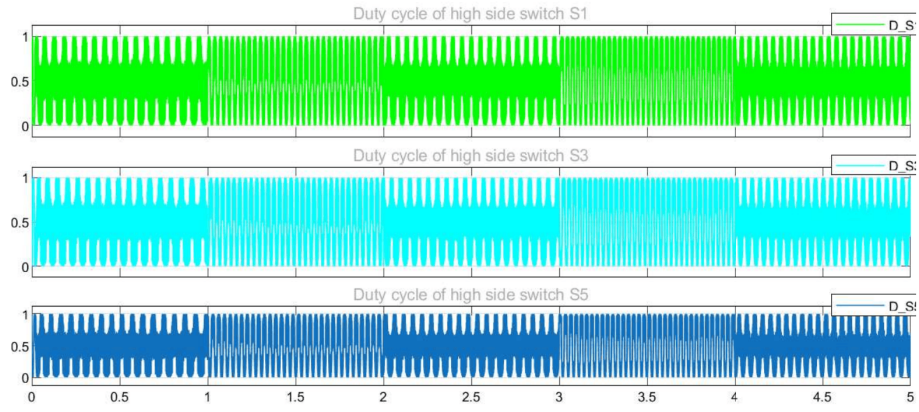


Figure 5.10: Duty Cycles for High-Side Inverter Switches (S_1, S_3, S_5)

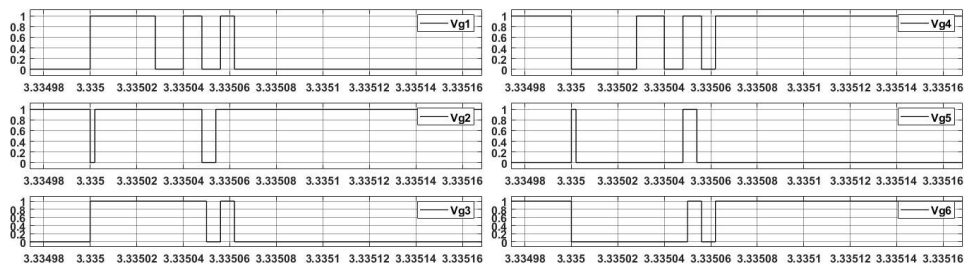


Figure 5.11: Final Discretized Gate Pulses for IGBTs

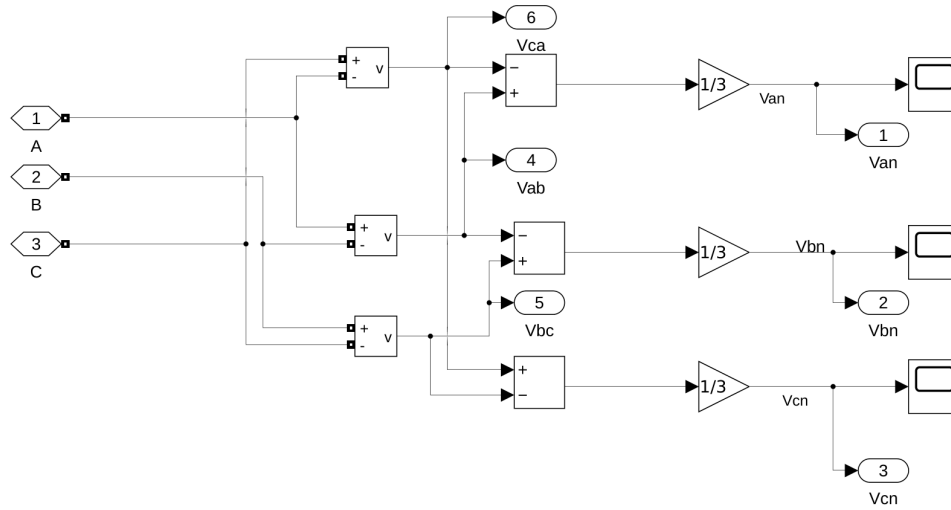


Figure 5.12: Phase and Line Voltage Measurements on the Plant

Ultimately, this Simulink model serves as an instructional framework, unifying electromagnetic machine theory, advanced discrete-time control systems, and power electronic actuation.

5.2 Future Scope

The simulation foundation established in this report supports several future implementations. Work will focus on transitioning this model to physical hardware via Hardware-in-the-Loop (HIL) testing and deployment using a microcontroller such as the STM32 framework. Furthermore, integrating sensorless control estimators to replace the physical encoder would support system reliability, while the application of Model Predictive Control (MPC) could be investigated to address the limitations of traditional PI regulators.

Bibliography

1. Texas Instruments, *Sensorless Field Oriented Control of 3-Phase Induction Motors*, Application Report SPRABP9, 2013.
2. Bose, B. K. *Modern Power Electronics and AC Drives*. Prentice Hall PTR, 2001.